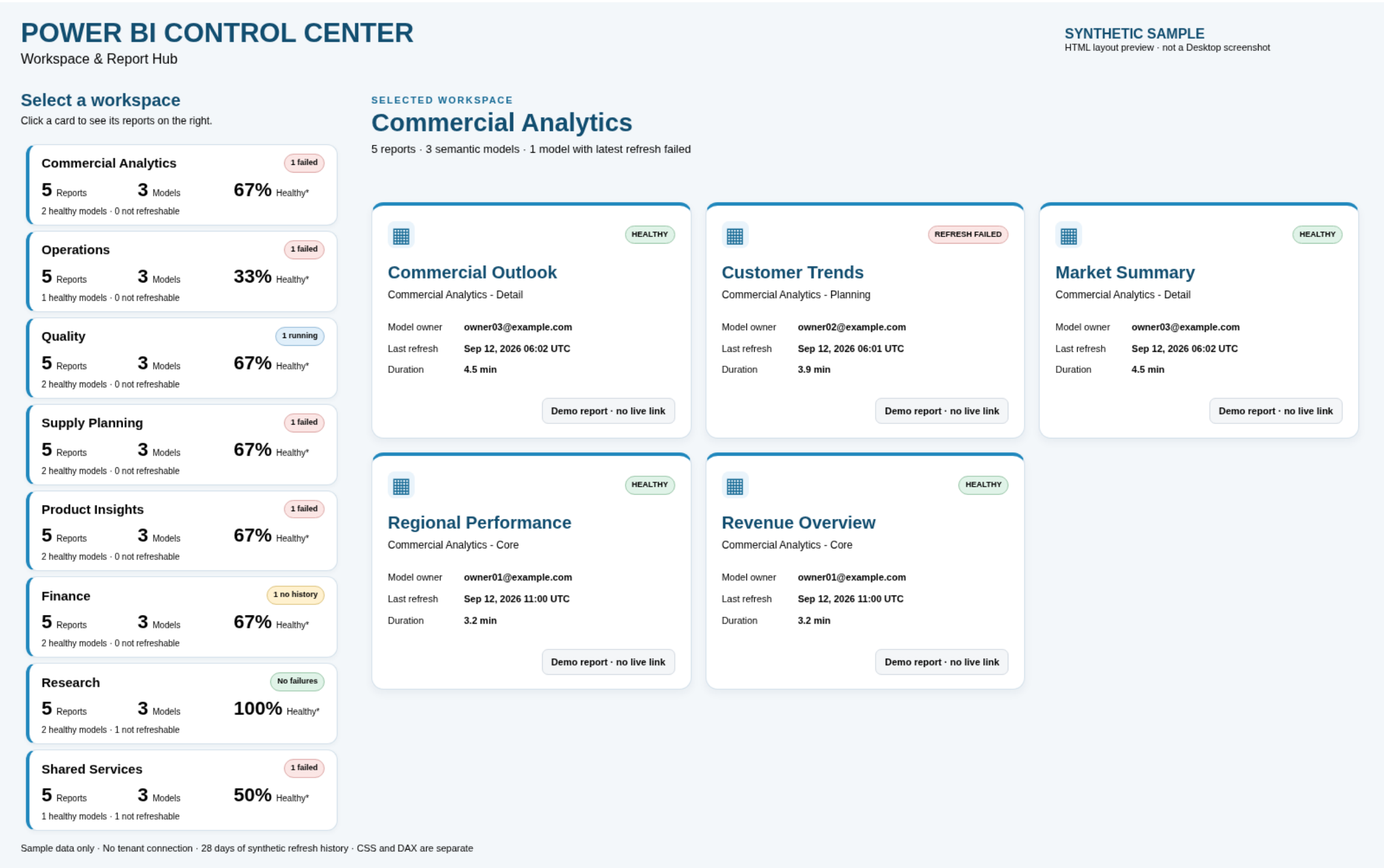


A practical control layer for the Power BI estate.

Built with the Power BI REST APIs, PowerShell, Power Automate, SharePoint, DAX and HTML/CSS.



The problem was not another dashboard.

It was visibility across a distributed Power BI environment.

- 01 What workspaces and reports exist?

- 02 Which semantic models support them?

- 03 What refreshed, what failed, and what has no history?

- 04 Can a user move from the overview directly to the report?

I wanted the metadata collection to be repeatable, the history to be retained, and the interface to feel like a control surface rather than an inventory spreadsheet.

The architecture we actually used.



POWER BI SERVICE

REST APIs expose workspace, report, model and refresh metadata



POWERSHELL

Calls APIs, handles pagination/errors, normalizes and enriches the response



SHAREPOINT

Stores CSV/Excel extracts and retained snapshot history



POWER BI MODEL

Relationships + DAX convert metadata into operational state



HTML / CSS

Renders the workspace hub, report details and health/status experiences

AUTOMATION

Power Automate schedules / moves metadata captures where appropriate.

PowerShell is the metadata collector.

It turns REST responses into a repeatable set of operational tables.

```
$paths = @{
  Workspaces    = "PBI_ControlCenter_Workspaces.csv"
  Reports       = "PBI_ControlCenter_Reports.csv"
  SemanticModels = "PBI_ControlCenter_SemanticModels.csv"
  RefreshHistory = "PBI_ControlCenter_RefreshHistory.csv"
  Current       = "PBI_ControlCenter_Current.csv"
  SnapshotHistory = "PBI_ControlCenter_SnapshotHistory.csv"
}

Completed -> HEALTHY
Failed    -> REFRESH FAILED
No history -> NO REFRESH HISTORY
```

API COLLECTION

- Workspaces
- Reports
- Semantic models
- Refresh history

PROCESSING

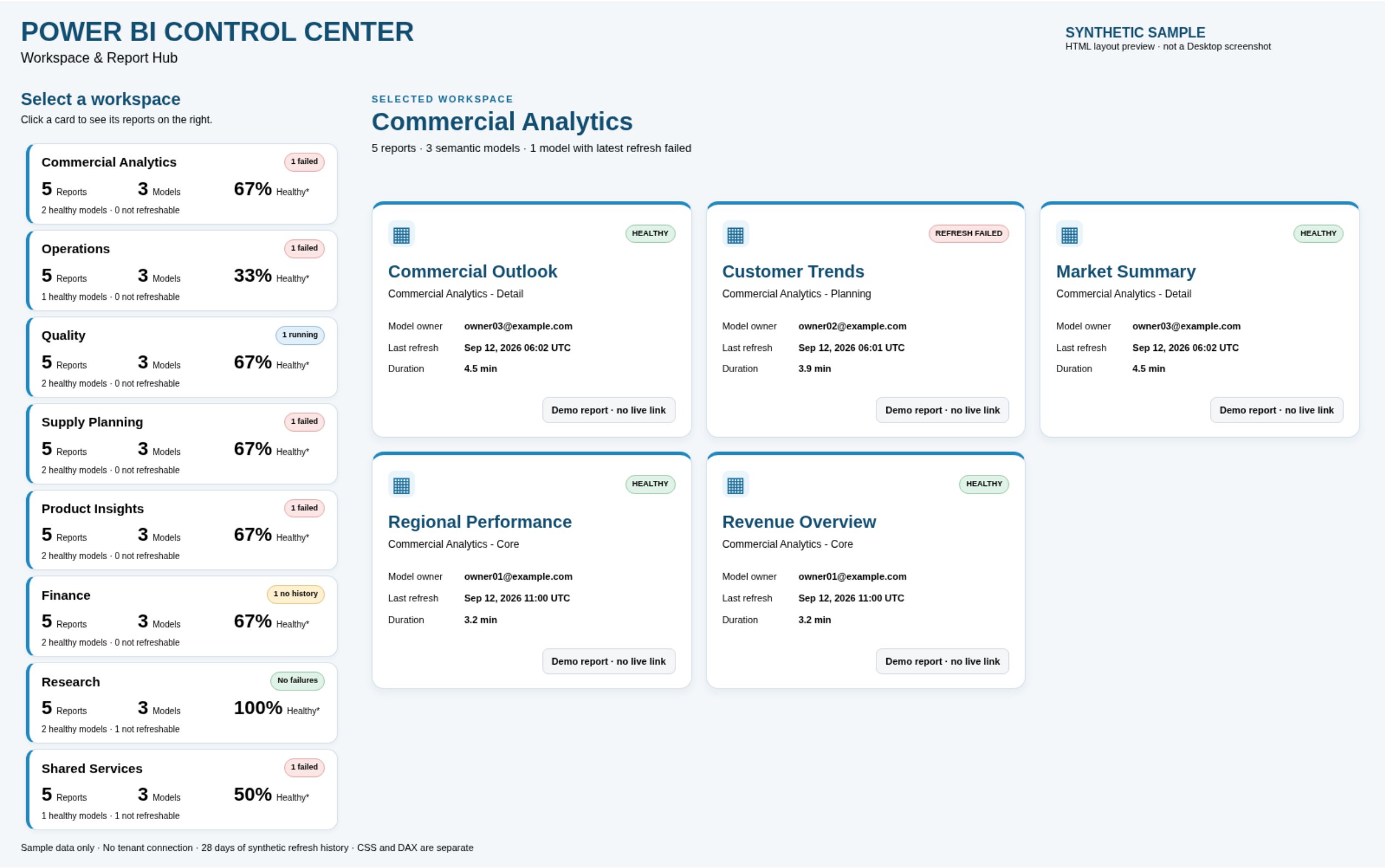
- Pagination
- Error capture
- Latest refresh lookup
- Health-state enrichment

OUTPUT

- Current state
- Snapshot history
- Run information
- CSV / Excel extracts

Workspace and report hub.

The model supplies the state. HTML/CSS controls how that state is presented.

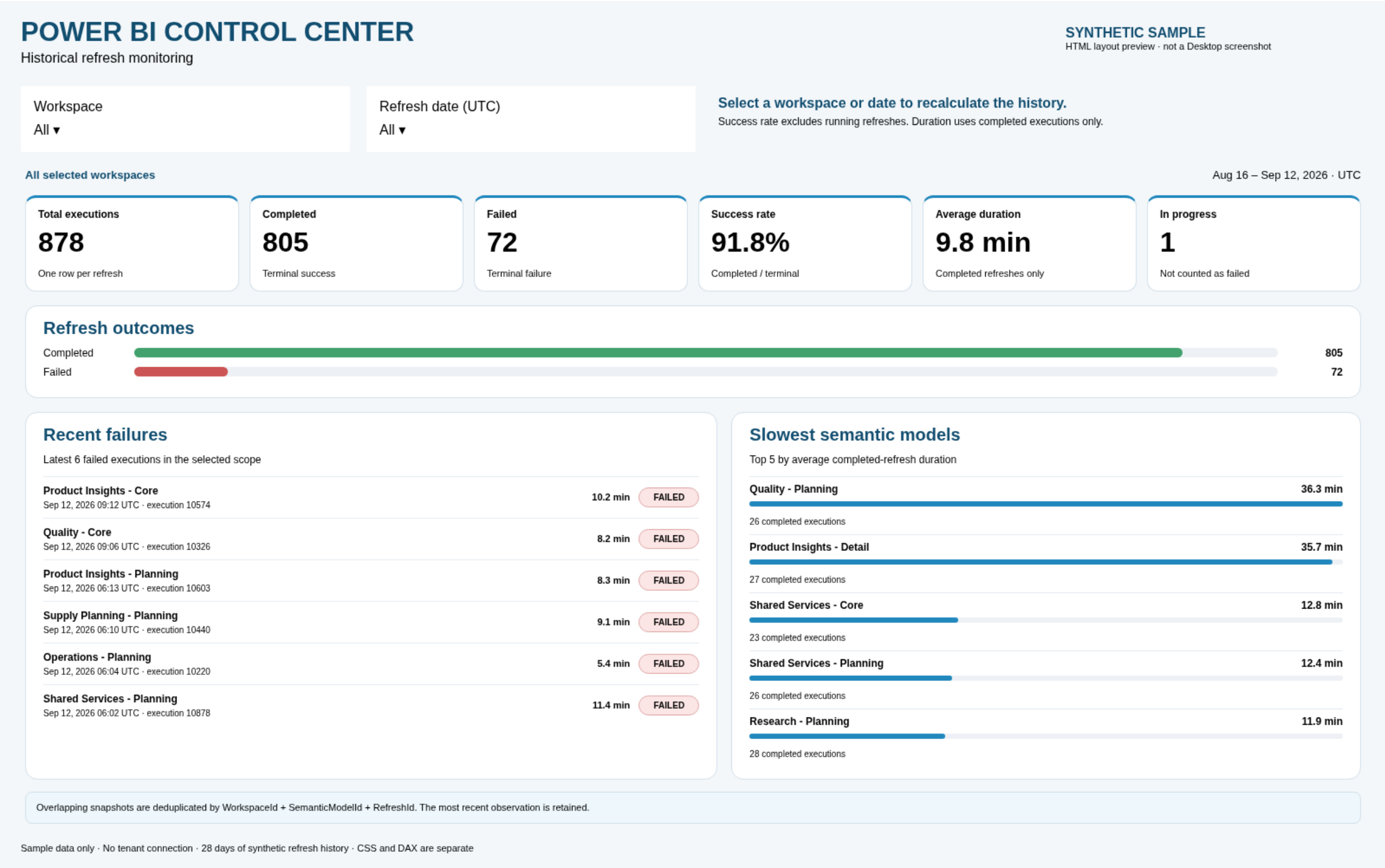


SYNTHETIC SAMPLE
HTML layout preview · not a Desktop screenshot

- 01 Workspace health + report/model counts 02 Selected workspace drives report detail
- 03 Healthy / failed / no-history state is visible without opening each report

Refresh history becomes operational context.

Current state and retained history answer different questions.



CURRENT

HISTORY

TREND

What is healthy right now?

What failed, when, and how often?

Which models are slow or unstable?

Power BI does not have to stop at charts.

REST APIs gave me the metadata.

PowerShell made collection repeatable.

SharePoint provided a lightweight history layer.

DAX turned metadata into operational state.

HTML/CSS gave me control over the experience.

THE POINT

Fabric would open additional possibilities.

This project was an exercise in seeing how far the Power BI ecosystem we already had could go.

Sanitized PBIP sample / synthetic data / DAX + CSS separated